***BRACT's* Vishwakarma Institute of Technology, Pune**

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No:1

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Subject Name: Deep Learning

Problem Statement:

To install and configure TensorFlow and Keras in Google Colab and perform the essential data preprocessing steps on the Fashion MNIST dataset. The experiment includes loading the dataset, exploring its structure, normalizing pixel values, splitting the data into training and testing sets (already provided in the dataset), and visualizing sample images to prepare the data for deep learning model development.

Algorithm:

1. Open Google Collab.
2. Import TensorFlow, Keras, NumPy, and Matplotlib libraries.
3. Verify the TensorFlow installation and version.
4. Load the Fashion MNIST dataset using Keras.
5. Display the shape of the training and testing datasets.
6. Visualize sample images from the dataset.
7. Normalize the pixel values by dividing them by 255.
8. Verify the normalized pixel value range.
9. Display the class labels corresponding to the images.
10. The dataset is now preprocessed and ready for model training.

Code & Output:

```
[1]
✓ 7s
# 1. Import Libraries

import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import fashion_mnist

print("TensorFlow Version:", tf.__version__)

TensorFlow Version: 2.20.0

[2]
Start coding or generate with AI.

Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz
29515/29515 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz
26421880/26421880 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz
5148/5148 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz
4422102/4422102 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Training Shape: (60000, 28, 28)
Testing Shape: (10000, 28, 28)

[5]
✓ 0s
# 3. Normalize Pixel Values (0-255 -> 0-1)

train_images = train_images / 255.0
test_images = test_images / 255.0
```

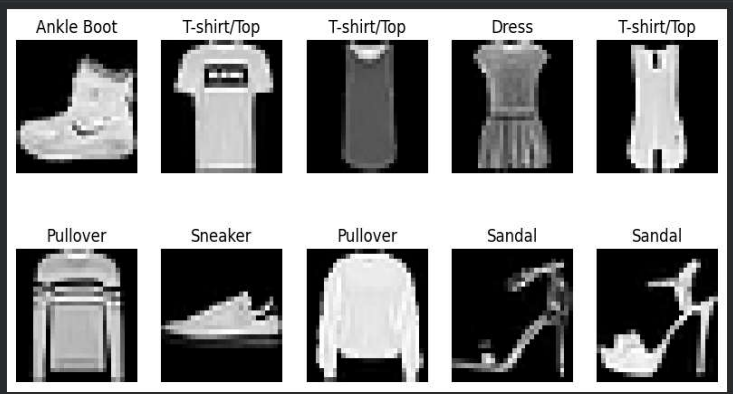
```
[6]
✓ 0s
# 4. Visualize Dataset

class_names = [
    'T-shirt/Top', 'Trouser', 'Pullover', 'Dress',
    'Coat', 'Sandal', 'Shirt', 'Sneaker',
    'Bag', 'Ankle Boot'
]

plt.figure(figsize=(10,5))

for i in range(10):
    plt.subplot(2,5,i+1)
    plt.imshow(train_images[i], cmap='gray')
    plt.title(class_names[train_labels[i]])
    plt.axis('off')

plt.show()
```



```python
# 5. Build Neural Network Model

model = tf.keras.models.Sequential([
    tf.keras.layers.Input(shape=(28,28)),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(10, activation='softmax')
])
```

```python
# 6. Compile Model

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

```python
# 7. Train Model

history = model.fit(
    train_images,
    train_labels,
    epochs=5,
    validation_split=0.2
)
```

```
Epoch 1/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.0996 - loss: 2.3027 - val_accuracy: 0.1005 - val_loss: 2.3027
Epoch 2/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.1041 - loss: 2.3023 - val_accuracy: 0.0957 - val_loss: 2.3020
Epoch 3/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 5s 3ms/step - accuracy: 0.1173 - loss: 2.3009 - val_accuracy: 0.1081 - val_loss: 2.2995
Epoch 4/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.1423 - loss: 2.2973 - val_accuracy: 0.2339 - val_loss: 2.2946
Epoch 5/5
1500/1500 ━━━━━━━━━━━━━━━━━━━━ 5s 4ms/step - accuracy: 0.1723 - loss: 2.2905 - val_accuracy: 0.1751 - val_loss: 2.2856
```

```python
# 8. Evaluate Model

test_loss, test_acc = model.evaluate(
    test_images,
    test_labels
)

print("Test Accuracy:", test_acc)
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.1725 - loss: 2.2856
Test Accuracy: 0.17249999940395355
```

```python
# 9. Make Predictions

predictions = model.predict(test_images)

print("Predicted Class:",
      class_names[predictions[0].argmax()])

print("Actual Class:",
      class_names[test_labels[0]])
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step
Predicted Class: Sandal
Actual Class: Ankle Boot
```

```python
# 10. Save Model

model.save("fashion_mnist_model.keras")
```